

Luis Ramirez

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Plainfield, NJ 07060



EDUCATION

New Jersey Institute of Technology (NJIT)

Newark, NJ

Bachelor of Science in Mechanical Engineering

GPA 3.24 — Expected May 2027

- Active member, Educational Opportunity Program (EOP)

Jun 2023 – Present

SKILLS & COURSEWORK

Technical & Professional Skills: Engineering drawing interpretation, assembly modeling, technical documentation, dimensional inspection, analytical problem solving, troubleshooting, teamwork, leadership, fluent bilingual Spanish
Software: SolidWorks & Creo Parametric (3D modeling, assemblies, drawings, BOMs), MATLAB, Microsoft Office
Relevant Coursework: Machine Design, Computer-Aided Design, Stress Analysis, Thermodynamics, Fluid Mechanics

PROJECTS

Modular Swing Frame Assembly | *SolidWorks*

Sep 2025 – Dec 2025

- Modeled a 10+ component structural assembly using fully constrained mates and parametric features; produced fabrication drawings, exploded views, and a complete BOM.
- Performed static FEA stress and deflection analyses to evaluate load-carrying capacity and identify areas for structural improvement.

Mechanical Roller Chain & Sprocket Drive | *Creo Parametric*

Sep 2025 – Dec 2025

- Designed a 20+ component mechanical drive assembly, generating 3D parts, exploded views, and manufacturing-ready drawings to support assembly and fabrication.
- Conducted kinematic motion simulations and tolerance stack-up checks to evaluate component interaction and ensure interference-free operation.

EXPERIENCE

Automotive Parts Specialist

Jul 2025 – Present

AutoZone Auto Parts

Newark, NJ

- Perform hands-on installation, testing, and troubleshooting of automotive components including 50+ batteries, 100+ lighting components, wipers, fuses, spark plugs, fluids, and air/cabin filters using battery testers, OBD-II scanners, torque tools, and other automotive equipment, while explaining findings and corrective actions.
- Apply vehicle-specific specifications including year, make, model, engine, trim, and part numbers to determine component fitment and diagnose issues; inspect for wear, leakage, overheating, and improper installation.

Field Engineering Technician

Aug 2021 – Apr 2023

Pittenger Paving & Excavating, Inc.

Dunellen, NJ

- Measured and verified 20+ construction sites using tape measures and laser-level equipment to confirm layout accuracy and structural tolerances, supporting proper installation of infrastructure components and execution.
- Operated heavy machinery including skid-steers, rollers, and tampers while assisting with paving and excavation projects, maintaining safety procedures and supporting efficient material placement in demanding environments.

LEADERSHIP & ACTIVITIES

EOP STEM Tutor

Jul 2024 – May 2025

New Jersey Institute of Technology (NJIT)

Newark, NJ

- Delivered 200+ hours of academic tutoring to first-year students in calculus, physics, and engineering, breaking down complex problem sets and exam preparation step by step.
- Facilitated 2-hour group study sessions and workshops 2–3 times per week in-person and via Discord, adapting problem-solving methods to individual student needs.

Alpha Phi Omega (APO) — Professional Service Fraternity

Sep 2025 – Present

Active Brother (NJIT)

Newark, NJ

- Completed 60+ hours of hands-on community service, including local animal shelter support in Montclair, community blood drives, and clothing donations.
- Collaborated with active brothers across 30+ chapter meetings to plan events, manage volunteer shifts, and support local Newark and regional outreach initiatives.